

UNIVERSITY EXAMINATIONS
COURSE CODE: COM 418
COURSE TITLE: EXPERT SYSTEMS
DURATION: 2 weeks
INSTRUCTIONS:

1. This is take away cat examination.
2. Answer **all** questions.
3. Plagiarism or direct copying from online sources without context will be penalized. Focus on application and justification.
4. Submit on 11th December 2025
5. Use A4 paper. Write a question plus its marks before answering it
6. All work should be handwritten

This set covers **100% of the syllabus**. Each question is worth **20 Marks** and contains a mix of **Short Answer (Theory)** and **Application (Scenario)** to meet all your requirements.

TOPIC 1: BASIC CONCEPTS AND TECHNIQUES

Covers: Data/Info/Knowledge, Heuristics, Advantages of ES, Production Systems.

Question 1

1. Define "**Heuristic**". How does heuristic search differ from algorithmic search? (4 marks)
 2. List **four** major components of a typical Expert System structure. Briefly explain the function of the **User Interface**. (4 marks)
- **(Application - 12 Marks):**
 - **Scenario:** You are explaining Expert Systems to a skeptical doctor.
 - Compare a human specialist and an Expert System. List **three** areas where the Human is superior and **three** areas where the System is superior.
 - Based on this comparison, would you recommend the system *replace* the doctor or *assist* the doctor? Justify.

Question 1B

1. Distinguish between **Data**, **Information**, and **Knowledge** using a "Traffic Light" example. (3 marks)
 2. What is a **Production System**? Define the roles of the "Rule Base" and "Working Memory". (3 marks)
- **(Application - 14 Marks):**
 - **Scenario:** A factory wants to automate machine repairs.
 - Write **four** production rules (IF-THEN) that represent the following knowledge: *"If the red light is blinking, check the oil. If the oil is low, add oil."*

If the oil is high but the light is still blinking, call a supervisor. If the green light is on, the machine is working."

- Does this represent **Procedural Knowledge** or **Declarative Knowledge**? Explain why.

Question 1C

1. Explain the concept of "**Explanation Facility**" (Justification). Why is it critical for user trust? (5 marks)
 2. Define **Domain Specificity**. Why can't we just build one "Super Expert System" to solve all problems (Medical, Legal, and Mechanical) at once? (5 marks)
- **(Application - 10 Marks):**
 - **Task:** Perform a **Suitability Assessment**.
 - Case 1: "Calculating the payroll for 1,000 employees."
 - Case 2: "Diagnosing rare skin diseases based on visual symptoms."
 - Which of these two is suitable for an Expert System and which is better for a standard algorithm? Explain your reasoning using the criteria of **Complexity** and **Structure**.

TOPIC 2: KNOWLEDGE REPRESENTATION & INFERENCE

Covers: Forward/Backward Chaining, Semantic Nets, Frames, Rules.

Question 2A (Chaining Architectures)

1. Define **Forward Chaining** (Data-Driven) and **Backward Chaining** (Goal-Driven). (4 marks)
 2. What is **Conflict Resolution**? If two rules are true at the same time, name one strategy to decide which to fire (e.g., Recency, Specificity). (4 marks)
- **(Application - 12 Marks):**
 - **Scenario:** Travel Agent System.
 - **Rules:**
 - R1: IF Ticket < \$500 THEN Class = Economy.
 - R2: IF Class = Economy AND Flight_Time > 5hrs THEN Serve = Snack.
 - R3: IF Class = Economy AND Flight_Time < 2hrs THEN Serve = Nothing.
 - **Facts:** Ticket = \$400, Flight_Time = 6hrs.
 - **Task:** Trace the inference. Show the steps to determine what to Serve. Is this Forward or Backward chaining?

Question 2B (Structural Representation)

1. What is a **Semantic Network**? What do the "Nodes" and "Links" represent? (3 marks)
 2. Explain "**Inheritance**" in the context of a Semantic Network. (3 marks)
- **(Application - 14 Marks):**
 - **Task:** Design a knowledge structure for a **University Library**.
 - Draw (or list nodes/links for) a Semantic Network connecting: *Book, Novel, Textbook, Harry Potter, Biology101, Author, JK Rowling*.
 - Use links like: *IS-A, WRITTEN-BY, INSTANCE-OF*.
 - Show how *Harry Potter* inherits the property *Has-Pages* from the concept *Book*.

Question 2C (Frames and Logic)

1. Compare **Propositional Logic** vs. **Predicate Logic**. Which is more powerful for Expert Systems? (4 marks)
 2. Describe a **Frame-based** representation. Define "Slots", "Facets", and "Demons". (4 marks)
- **(Application - 12 Marks):**
 - **Scenario:** Real Estate System.
 - Create a **Frame** for a generic "House".
 - Include Slots for: *Number_of_Rooms, Location, Price, Type*.
 - Create an **Instance Frame** for "Villa Rosa" that inherits from "House" but overrides the *Type* to "Mansion". Explain how "Default Values" work in this context (e.g., if Rooms is unknown, assume 3).

TOPIC 3: UNCERTAINTY MANAGEMENT

Covers: Probability, Fuzzy Logic, Certainty Factors.

Question 3A (Sources and Fuzzy Logic)

1. Identify **three** sources of uncertainty in data. Give an example of each. (6 marks)
 2. What is the primary difference between **Boolean Logic** (Crisp) and **Fuzzy Logic**? (2 marks)
- **(Application - 12 Marks):**
 - **Scenario:** Automatic Washing Machine.
 - Variables: *Dirt_Level* (Low, Medium, High).
 - If a shirt has 45% dirt, Boolean logic says it is "Not Dirty". Fuzzy logic might say it is "0.4 Dirty and 0.6 Clean".
 - Explain how **Fuzzy Membership Functions** would handle a shirt that is "Slightly Dirty". Draw a rough graph showing the overlap between "Clean" and "Dirty".

Question 3B (Certainty Factors)

1. What does a **Certainty Factor (CF)** of +1.0, 0.0, and -1.0 represent? (3 marks)
 2. Why do we use CFs in medical diagnosis instead of strict True/False? (3 marks)
- **(Application - 14 Marks):**
 - **Scenario:** Weather Prediction.
 - Rule: IF Humidity is High AND Temp is Drop THEN Rain is Likely (CF = 0.9).
 - **Observation:** The sensor is 100% sure Humidity is High (CF=1.0). The sensor is only 80% sure Temp is Drop (CF=0.8).
 - **Calculation:**
 1. Find the CF of the **Premise** (Minimum of the inputs).
 2. Multiply by the Rule Strength (0.9).
 3. What is the final confidence in "Rain"?

Question 3C (Bayesian vs. Approaches)

1. Briefly explain **Bayesian Reasoning**. It relies on "Prior Probability" and "Posterior Probability" - define these terms. (6 marks)
 2. Name one disadvantage of using Bayesian probability in large expert systems (Consider the amount of data required). (4 marks)
- **(Application - 10 Marks):**
 - **Task:** You are building a system to detect email spam.
 - The phrase "Free Money" appears. In non-spam, this phrase appears 1% of the time. In spam, it appears 50% of the time.
 - Explain conceptually how the system updates its belief that an email is spam after seeing this phrase. (You do not need to perform the full math, just explain the logic of updating belief).

TOPIC 4: KNOWLEDGE ENGINEERING & BUILDING SYSTEMS

Covers: Development Life Cycle, Knowledge Acquisition, Verification.

Question 4A (The Development Cycle)

1. List the generic phases of Expert System Development. (4 marks)
 2. Why is the **Waterfall Model** generally considered unsuitable for Expert Systems development? (4 marks)
- **(Application - 12 Marks):**
 - **Scenario:** You are the Project Manager for an Agricultural Pest Advisor system.
 - You have finished the **Initial Prototype**.

- Describe the **Field Testing / Validation** process. Who should test it? How do you measure "Success"? (e.g., comparison against a human agronomist). What happens if the system makes a mistake?

Question 4B (Knowledge Acquisition)

1. Define the "**Knowledge Acquisition Bottleneck**". (3 marks)
 2. Who is the **Knowledge Engineer**? How does their role differ from the **Domain Expert**? (3 marks)
- **(Application - 14 Marks):**
 - **Scenario:** You are extracting rules from a veteran Car Mechanic who "just knows" what is wrong by listening to the engine.
 - He cannot articulate his rules easily (Tacit Knowledge).
 - Propose **two** specific elicitation techniques (e.g., Protocol Analysis, Interview, Repertory Grid).
 - Describe exactly how you would apply *each* technique in this garage setting to generate IF-THEN rules.

Question 4C (Building a Small System)

1. What is a **Feasibility Study**? Define Technical, Economic, and Social feasibility. (6 marks)
 2. What does it mean to "prune" the knowledge base? (2 marks)
- **(Application - 12 Marks):**
 - **Task:** Conceptualize a small ES for "**Choosing a Pet**".
 - Define the **Goal**: (Recommending a Dog, Cat, or Hamster).
 - List **three** key questions the system must ask the user (Inputs).
 - Write the logic (Pseudocode/Rules) that connects these inputs to the three recommendations.

TOPIC 5: TOOLS, MARKET & SIGNIFICANT SYSTEMS

Covers: Shells, History (MYCIN/DENDRAL), Modern Market.

Question 5A (Tools and Shells)

1. Define an **Expert System Shell**. Give one example. (4 marks)
 2. List two advantages and two disadvantages of using a Shell versus programming in Python/C++. (4 marks)
- **(Application - 12 Marks):**
 - **Scenario:** A small startup wants to build a legal advisor bot. They have limited funds and need a working demo in 2 weeks.
 - Would you recommend they use a **Shell** or a **Programming Language (LISP/PROLOG)**?

- Justify your choice based on: *Speed of development*, *Cost*, and *Flexibility*.

Question 5B (Significant Systems & History)

1. Briefly describe **MYCIN**. What was its domain? What contribution did it make to AI (4 marks)
 2. Briefly describe **PROSPECTOR**. What was its domain? (2 marks)
 3. What was the "**AI Winter**" of the late 1980s? (2 marks)
- **(Application - 12 Marks):**
 - **Analysis:** Why did systems like MYCIN never become products used in actual hospitals, despite being accurate?
 - Discuss the **Integration** issues (connecting to hospital records) and **Liability** issues (who is sued if the patient dies?) that prevented their market success.

Question 5C (The Expert System Market)

1. How has the "Expert System" market changed in the 21st century? (Hint: Embedded systems vs Standalone). (4 marks)
 2. What is **Data Mining**? How does it relate to modern rule discovery compared to manual knowledge engineering? (4 marks)
- **(Application - 12 Marks):**
 - **Scenario:** A bank uses an AI system to approve loans.
 - **Ethical Audit:** The system starts rejecting loans for people from a specific postcode.
 - Analyze this from a **Social Factor** perspective. Is the system "Biased"? How did the bias likely get into the Knowledge Base (Training Data vs Rules)?
 - How would you fix this?